Zewail City of Science, Technology and Innovation

University of Science and Technology

School of Computational Sciences and Artificial Intelligence

GRADUATION PROJECT

MARKET VISIBILITY REPORT

FarmPulse AI

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1. Executive Summary

On most Egyptian cattle fattening farms, workers still check animals by eye a few times a day. Slow changes in how a cow moves, eats, lies down or breathes are easy to miss, and nobody keeps a history for each animal that shows what is normal for it. FarmPulse AI uses fixed barn cameras to detect and follow every cow, recognise its behaviour and visible signs such as open mouth breathing and rumination, and build a dated record for each animal. When a cow drifts away from its own baseline, the system sends a clear alert with the video clip, and a farmer or vet decides what to do. The same cameras also raise an immediate alert for intruders.

Recent studies (2024 to 2026) show that detection, tracking and behaviour recognition already work well, but each one is tested alone, on one farm and short clips, and almost never against real veterinary outcomes. Commercial products either need a sensor on every animal, like SenseHub Feedlot and smaXtec, or watch for a single problem, like CattleEye. None of them offers tag free health monitoring of each cow in housed fattening pens. FarmPulse AI is built to fill exactly that gap.

2. Problem Statement & Motivation

State the specific problem clearly and, where possible, quantify it (cost, error rate, time, efficiency loss). Explain who is affected, why existing solutions are insufficient, and why this problem matters now (technical, economic, or social significance). Precisely defines the real world computing problem, thoroughly analyzes impact on stakeholders, and provides compelling motivation for immediate solution.

2.1 Core Problem Definition

Fattening farms depend on periodic visual checks, so early signs of illness in individual cows are often missed and there is no history for each cow to act on.

2.2 Problem Context & Real World Impact

Egypt had about 2.81 million cattle in 2021 and imported around $256 million of live animals that year against $4 million of exports, while the state aims for self sufficiency in meat [1]. On a fattening farm, every sick animal caught late means lost weight gain, higher treatment costs or a death, and all of it comes straight out of the farm's margin.

Owners carry these losses, workers spend hours walking the pens, and vets are usually called only once the signs are obvious, with no record of how long the change has been building. Manual checking is also heavy on labour: in one feedlot trial, riders who checked pens by eye covered 18 pens a day, while riders guided by a sensor system only needed to visit 5.3 [2].

2.3 Motivation & Urgency

Wearable sensors show that illness can be flagged up to four days before clinical signs appear [3], which is exactly the window a periodic check misses. At the same time, plain cameras and edge computing have become cheap, and about 80% of recent dairy computer vision studies already rely on ordinary RGB cameras [4].

Yet the products on the market are built for large dairy herds in Europe and North America and depend on imported hardware fitted to every animal, which cost, connectivity and skills gaps make hard to adopt in Africa [5]. A camera only system designed with local fattening farms is both possible today and missing from the market.

3. Proposed Solution & Program Focus

Describe your computing based solution, key capabilities, and track specific technical details aligned with your program focus. Presents a comprehensive computing solution addressing the problem; demonstrates strong technical feasibility and program specific depth (SWD architecture, IT integration/security, or DSAI pipeline/models).

3.1 Solution Overview & System Workflow

FarmPulse AI turns continuous barn video into a health record for each cow. Cameras in every pen stream time stamped video. A detector finds cows, people and vehicles, and a tracker with reidentification gives each cow a stable ID. Three vision models then run on the tracked animals: behaviour (standing, lying, walking, feeding, drinking), visible state (open mouth, rumination) and intruder detection.

Observations are saved to a cow's record only when its identity is reliable; anything uncertain stays as a zone level event instead of being pinned on the wrong animal. Simple veterinary rules compare each cow with its own baseline and with the rest of the pen, and raise a risk alert with the supporting clip. The farmer or vet confirms or dismisses it, and that feedback becomes the outcome labels for later models. Intruder events go out at once as a separate security alert, and a weekly report and dashboard summarise the herd.

3.2 Key System Features

Feature 1: Tag free monitoring of every cow. Cameras recognise and follow each animal

without ear tags, collars or boluses, so the farmer can see each cow's behaviour history over days and weeks.

Feature 2: Explainable health alerts. Every alert says which signal triggered it and comes with the video clip, so a farmer or vet can check it in seconds and record what happened.

Feature 3: Security alerts, weekly reports and a dashboard. The same cameras flag intruders right away, and a weekly report and simple dashboard show herd risk and flagged animals at a glance.

3.3 Track Specific Technical Focus

Detail technical elements based on your program track:

- **SWD (Software Development):** Software architecture, design patterns, core algorithms, module breakdown, and frameworks.
- **IT (Information Technology):** Systemic integration, network topology, security mechanisms, cloud deployment, and system administration protocols.
- **DSAI (Data Science & AI):** Data collection lifecycle, preprocessing pipelines, feature engineering, ML/DL models, and evaluation metrics.

Data collection: We record day and night video at our two partner farms (Al Ammar and Al Tawam) with farm, camera and zone metadata, cut it into versioned clips and label them with a written guide. A second annotator or the vet reviews a subset, and we report agreement as Cohen's kappa. Health outcomes come from the vet's feedback on alerts.

Preprocessing: Frame sampling, resizing, low light enhancement for night footage, augmentation and pen masks. Data is split by cow and by recording period, so near identical frames never leak between training and testing.

Feature engineering: For each cow we compute daily time budgets (lying, feeding, standing, walking), rumination bouts and open mouth episodes, plus how far each value drifts from the cow's own rolling baseline and from the pen median.

Models: YOLOv8 or YOLOv11 detectors for cows, people and vehicles; DeepSORT or OC SORT tracking with a ResNet18 reidentification network; a clip level video model for behaviour and rumination; a small object detector for the mouth region; rule based risk scoring first, then a gradient boosted classifier once enough verified outcomes exist; and an offline reinforcement learning study that compares a suggested response policy with a fixed one.

Evaluation: Precision, recall and F1 per class with confusion matrices; mAP for detection; MOTA, IDF1, HOTA and ID switches for tracking; and for alerts, recall, false alert rate and lead time on unseen cows and periods, with thresholds agreed with the vet.

4. Target Audience & User Persona Analysis

Demonstrate a clear understanding of who will use your product, their characteristics, and why your solution is a strong fit. Identifies specific user personas and usage contexts; clearly articulates user needs and justifies the value delivered by the solution.

4.1 Primary User Persona

Owners and managers of small and medium beef fattening farms in Giza and nearby governorates. A typical operation finishes around 100 head a year [1] with a few workers and no technical staff. The owner uses a smartphone and WhatsApp every day but is not a technical user, and profit depends on daily weight gain and avoiding losses. Secondary users are the farm vet, who reviews alerts, and the workers, who carry out pen checks.

4.2 User Characteristics, Goals, and Usage Context

Their goals are simple: spot sick animals early, cut treatment costs and deaths, use staff time well and keep animals safe from theft. The pain points are that pens cannot be watched day and night, small changes slip between checks, there is no history for each cow, and imported sensor systems are expensive and made for dairy. They work in dusty, semi open barns with changing light and sometimes weak internet, read alerts on a phone in Arabic, check a dashboard once or twice a day and go through the weekly report with their vet.

4.3 User Need & Value Proposition

These farms need eyes on every animal around the clock, which their staff cannot provide. FarmPulse AI sends workers only to the cows that need attention, backs every alert with a clip they can trust, gives the vet a behaviour history for earlier and better decisions, and adds theft protection from the same cameras. Because nothing is fitted to the animals, the cost grows with the number of pens, not the number of cows.

5. Literature Review & Background Research

Survey the related research that underpins your project. Summarize 5 to 10 key works (peer reviewed, preferably from the last 5 years): what was done, the methods and data used, the results, and the limitations. Analyze critically rather than descriptively, and conclude with the research gap your project addresses.

5.1 Background & Key Concepts

Object detectors such as YOLO find animals in each frame and are scored with mAP. Multi

object tracking links those detections over time using a motion model plus appearance features, and is scored with MOTA, IDF1, HOTA and ID switches. Reidentification learns what each cow looks like so it can be recognised again after occlusion or in another camera. Behaviour recognition classifies short clips into actions such as feeding or ruminating, and pose estimation tracks body keypoints to measure gait.

On the veterinary side, changes in feeding, lying and rumination time, lameness and open mouth breathing are well known signs of illness or heat stress. Risk can be scored with transparent rules that need no labels, or with a trained classifier that needs verified outcomes; in both cases a person makes the final call.

5.2 Summary of Related Work

Ref.	Author(s), Year	Method / Approach	Dataset / Context	Key Results	Limitations
[4]	Antognoli et al., 2025	Literature review of computer vision in dairy farm management	92 documents from 2014 to 2024 in Scopus, Web of Science and PubMed	About 80% of studies use plain RGB cameras; strong results for identification, feeding, body condition and lameness	Dairy only; most reviewed work comes from single farms with little validation across farms
[5]	Tadele et al., 2025	Review of precision dairy farming in Africa	African dairy systems, smallholder and commercial	Cost, connectivity, digital skills and the lack of locally adapted tools are the main barriers	Qualitative; no measured barrier figures; dairy only
[6]	Li K. et al., 2024	SlowFast (ResNet50) baseline on a	107 dairy cows, 7 cameras,	mAP about 71 to 74%; standing, lying	Rumination only 41% AP and drinking

Ref.	Author(s), Year	Method / Approach	Dataset / Context	Key Results	Limitations
		new cow behaviour video dataset	687 clips	and feeding above 96% AP	35%; behaviours not linked to cow IDs; one farm
[7]	Liu et al., 2025	CLIP vision transformer adapted to video with custom prompts and few shot training	1,905 clips, six behaviours, 200 cow dairy farm	96.1% accuracy; ruminating F1 0.94 to 0.97	Lying ruminating confused with grooming; one farm; preprint
[8]	Guarnido-Lopez et al., 2024	YOLOv8m vs YOLOv10m for feeding activity	Charolais beef bulls on a French farm, 1,200 frames	mAP 0.94; biting and chewing F1 0.91 to 0.99	Visiting recall only 0.15 to 0.37; small single farm dataset
[9]	Li Z. et al., 2025	Lightweight YOLOv11n with OC SORT and three other trackers	Commercial dairy farm, 7,380 images	MOTA 97.0%, IDF1 93.1%, 18.6% fewer parameters	Only two tracking videos; motion only matching; no edge device test
[10]	Yu et al., 2025	Supervised and self supervised reidentification across three cameras	101,329 images of 90 Holstein Friesian cows over 7 days	About 96% accuracy without manual ID labels	Known cows only; 57% in a crowded view; no behaviour analysis

Ref.	Author(s), Year	Method / Approach	Dataset / Context	Key Results	Limitations
[11]	Ma et al., 2025	Modified YOLOv8n for open mouth breathing	1,000 images of caged broilers, one farm	0.92 mAP50 on an independent test set	Chickens, not cattle; small dataset
[12]	Li G. et al., 2024	YOLOv8n with snake convolution and BiFormer attention; DeepSORT with ResNet18 reidentification	Beef farm in Anhui, China; 34,560 images of 9 behaviours; 45 cattle	93.6% behaviour detection; ID switches cut by 65.8%	Tracking tested on 5 short videos; one farm; assumes a fixed herd size
[13]	Sim and Cho, 2025	YOLOv9t detector with improved DeepSORT and StrongSORT (box width in the Kalman state, Pearson distance for matching)	4 cattle in one research pen, one camera	HOTA 78.6%, IDF1 91.4%; ID switches down from 18 to 5	Only eight 10 second clips; occlusion not fully solved; no behaviour
[14]	Fang et al., 2025	Benchmark of 28 detectors and 23 identification backbones	8 Holstein calves; one hour of annotated video, 537,908 boxes	Detection mAP50 95 to 99%; best identification only 73%	One of nine videos annotated; labels biased toward YOLO; preprint

Ref.	Author(s), Year	Method / Approach	Dataset / Context	Key Results	Limitations
[15]	Russello et al., 2024	Pose estimation of 9 keypoints turned into 6 gait traits for lameness	Outdoor videos of walking dairy cows	99.6% of keypoints correct; 80.1% lameness accuracy	Needs cows walking past a fixed camera; labels from observer scores
[16]	Zhang et al., 2024	EfficientNetV2 video features and BiLSTM collar data fused with a Transformer	5 Holstein cows, 228 labelled clips	98.8% accuracy, better than video or sensor alone	Five cows; needs a collar; no long term identity or health outcomes
[17]	Yu et al., 2026	Detect, segment (SAM) and identify pipeline with unsupervised contrastive reidentification	Nine days of CCTV on a working dairy farm	98.9% detection and 94.8% reidentification in dense crowds	One farm; relies on Holstein coat patterns; preprint
[18]	Delwar et al., 2025	YOLOv8 on IoT cameras with SMS alerts and a speaker deterrent	580 images of animal and human intruders on farms	Above 99% reported accuracy	Small dataset; built for crop farms; separate from animal health
[19]	Scott et al., 2024	Systematic review of vision methods	Large mammals including	Deep learning now dominates detection,	Outdoor focus; preprint; few quantified

Ref.	Author(s), Year	Method / Approach	Dataset / Context	Key Results	Limitations
		for outdoor livestock and wildlife monitoring	cattle, horses, sheep and deer	counting and species classification	comparisons

5.3 Critical Synthesis & Research Gap

Most works follow the same recipe: a YOLO style detector, a SORT style tracker and a learned appearance model [6], [8], [9], [12]. The trend is toward lighter models that can run on the farm [9] and self supervised reidentification that needs fewer labels [10]. Some results conflict in telling ways: rumination scores 41% AP in wide barn views [6] but above 0.94 F1 on close, curated clips [7], so benchmark numbers depend heavily on camera view and do not carry over directly to a whole pen.

The same limitations keep coming back. Almost every study uses one farm and one camera setup; tracking is judged on a handful of short clips [9], [12], [13]; reidentification is tested on Holstein cows with distinct coat patterns and drops sharply in crowds [10], [14], although a newer segmentation based pipeline recovers 94.8% in dense groups [17]; behaviour is recognised per frame or clip without a lasting cow ID [6], [7]; health work stops at observer scores [15]; and multimodal work relies on sensors worn by the cow [16]. Open mouth detection exists only for broilers [11].

The reviews confirm the picture: vision work is strong on single tasks but weak on generalisation across farms [4], [19], adoption in Africa is held back by cost and skills [5], and farm security is treated as a separate system built on small datasets [18].

The gap is clear: no study combines detection, tracking, reidentification, behaviour and visible state into a long term record for each housed beef cow using cameras alone, and then checks the resulting alerts against vet confirmed outcomes on unseen cows and periods. FarmPulse AI targets exactly that.

6. Market Visibility: Existing Solutions & Competitive Analysis

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Review similar systems and existing solutions (commercial products, open source tools and academic prototypes), compare them against your proposed solution, critically analyze their strengths and weaknesses, and identify the gaps your solution fills. Cover at least three alternatives where possible.

6.1 Review of Similar Systems & Existing Solutions

Solution / Provider	Type	Description & Core Functionality	Target Users	Availability / Pricing
SenseHub Feedlot (Merck Animal Health) [2]	Commercial	Ear tag that measures temperature and activity; machine learning flags animals that drift from normal and a light on the tag blinks	Beef feedlots	Tag on every animal plus antenna and gateway; price on request
CattleEye (GEA) [20]	Commercial	One overhead camera at the milking parlour exit scores lameness and body condition; over 100,000 cows monitored	Dairy farms	Sold within GEA dairy software; price not published
OneCup AI BETSY [21]	Commercial	Ranch cameras detect animals, people and vehicles, identify animals visually and flag limping, calving and feeding	Beef and dairy ranches	Price not published

6.2 Feature Comparison Matrix

Evaluation Feature	FarmPulse AI	A: SenseHub Feedlot	B: CattleEye	C: OneCup AI BETSY
Core Technology	Barn cameras with YOLO detection, tracking, reidentification	Ear tag sensors (temperature and accelerometer) with machine	Single 2D camera with gait analysis AI	Ranch cameras with computer vision

Evaluation Feature	FarmPulse AI	A: SenseHub Feedlot	B: CattleEye	C: OneCup AI BETSY
	and video behaviour models	learning		
Key Features	History for each cow, open mouth and rumination detection, alerts with clips, intruder alert, weekly report	Daily list of likely sick cattle and a blinking tag	Lameness and body condition scores	Animal, person and vehicle detection; limping and calving alerts
Performance / Accuracy	Precision and recall per class, alert recall, false alert rate and lead time on unseen cows (targets set with the vet)	Fewer respiratory disease deaths than visual checks in a feedlot trial [2]	No accuracy published	No accuracy published; weaker with mud or poor camera angle
User Experience	Arabic phone alerts with the clip, simple dashboard, a person approves every action	Simple pull list and a visual cue on the tag	Scores inside dairy herd software	Graphical alert app
Integration Capability	Works with ordinary IP cameras; one	Needs its own tags, antenna	Tied to GEA DairyNet	Proprietary platform

Evaluation Feature	FarmPulse AI	A: SenseHub Feedlot	B: CattleEye	C: OneCup AI BETSY
	video feed serves health and security	and gateway		
Platform & Availability	Prototype for Egyptian fattening farms; runs on site or in the cloud	Commercial, North America	Commercial, dairy markets	Commercial, Canada
Cost Structure	Grows with the number of pens; nothing fitted to animals	Tag per animal plus site hardware	Camera system per site	Not published

6.3 Strengths & Weaknesses Analysis

Competitor	Strengths	Weaknesses / Limitations	Implication for Your Solution
SenseHub Feedlot	Made for beef feedlots; proven drop in deaths and rider workload; very simple workflow	Device on every animal; tags get lost; only temperature and activity, with no visual evidence	Keep the simple pull list idea, avoid hardware on animals, and add a clip to every alert
CattleEye	Nothing on the cow; one camera; proven at scale	Only lameness and body condition; needs cows to walk past a fixed point, which fattening pens do not have	Keep the camera only approach, but watch cows inside the pen and cover more health signs

Competitor	Strengths	Weaknesses / Limitations	Implication for Your Solution
OneCup AI BETSY	Beef focused; visual ID without tags; also detects people and vehicles	No published accuracy; sensitive to mud and camera angle; built for open ranges	Combine health and security in the same way, but publish results per class and handle night footage in housed pens

6.4 Gap Analysis & Key Differentiators

#	Identified Gap	Evidence (Lit. / Competitor Ref.)	How Your Solution Addresses It
1	Health monitoring needs a device on every animal	[3], SenseHub Feedlot	Cameras only; cost scales with pens, not cows
2	Behaviour is not tied to a stable cow identity over time	[6], [7], [10], [12]	Records saved for a cow only when its ID is reliable, backed by reidentification
3	No open mouth breathing detector for cattle, and camera products cover a single condition	[11], CattleEye	Visible state model plus one risk score that combines behaviour, rumination and open mouth episodes
4	Alerts are a black box and rarely checked against vet outcomes	[15], SenseHub Feedlot	Transparent rules first, a clip with every alert, and a learned model only after vet confirmed outcomes

FarmPulse AI is the only option here that watches every housed fattening cow without fitting anything to it, links what it sees to each animal's own history, and backs every alert with a clip the farmer can check. It closes the research gap between isolated benchmarks and a validated monitoring workflow for each cow, and the market gap left by wearable and dairy focused products in price sensitive markets like Egypt.

7. Project Roadmap & Deliverables

Break the remaining project work into phases with clear dates, key activities and milestones, aligned with the GP lifecycle (Requirements/SRS: Week 10, Initial Design & Proof of Concept: Week 12, Draft Report: Week 13, Part 1 Defense: Week 15), followed by the Graduation Project 2 phases.

Project Timeline & Work Breakdown

#	Phase	Key Activities	Start / End	Milestone
1	**Market visibility and stakeholder input**	Finish the literature and market review; interview farmers and the vet about camera placement, alerts and budget	22 Oct to 4 Nov 2026	Market visibility report
2	**Requirements and data collection**	Write the SRS; install cameras; record day and night video; collect veterinary indicators; write the labelling guide	1 Nov to 25 Nov 2026	SRS (Week 10); first dataset version
3	**Initial design and proof of concept**	System architecture and database design; label first clips; baseline detector and tracker on farm video; one clip to alert demo	26 Nov to 9 Dec 2026	Design and PoC (Week 12)
4	**Draft report and Part 1 defense**	Baseline behaviour and intruder results on unseen cows; write the report; prepare the defense	10 Dec to 30 Dec 2026	Draft report (Week 13); Part 1 defense (Week 15)
5	**Graduation Project 2**	Train all vision and reidentification models; cow records and risk rules; alert feedback loop; dashboard and	Jan to Jun 2027	Working prototype, final report and defense

#	Phase	Key Activities	Start / End	Milestone
		weekly report; supervised risk model and offline RL study; final evaluation		

8. Risk Management & Professional Ethics

Identifies all plausible technical and project risks with actionable mitigation strategies; demonstrates professional conduct, honesty, and awareness of ethical/societal considerations.

Identified Risk	Likelihood / Impact	Mitigation Strategy
Occlusion and similar looking cows cause ID switches	High / High	Several camera angles per pen, appearance based reidentification, and records saved only above a track quality threshold
Night or low light footage	High / Medium	IR cameras, low light augmentation, and separate day and night evaluation
Too few confirmed illness cases to train a risk model	High / High	Start with vet rules, collect feedback on every alert, train only when there are enough labels and report it honestly if not
Rare signs such as open mouth or drinking are underrepresented	Medium / High	Targeted clip collection, balanced sampling and results reported per class
Delays in farm access or recording	Medium / High	Two partner farms, early camera installation, buffer weeks, and labelling while collection continues

Identified Risk	Likelihood / Impact	Mitigation Strategy
Vet availability for labels and reviews	Medium / Medium	A fixed weekly review slot and batch review through the dashboard
Weak internet on the farm	Low / Medium	Process video on site, queue alerts until the connection returns, and fall back to SMS

Professional ethics: FarmPulse AI supports decisions; it does not diagnose or treat, so a person approves every action and the RL study stays offline. We will report results per class, ID switches and false alerts honestly rather than a single accuracy number. Cameras may record workers, so we will get consent, limit person detection to security zones and hours, blur faces where we can and delete old footage on a schedule. Farm data belongs to the partner farms and is shared only with their permission, and data collection never involves touching or tagging the animals.

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Use a consistent citation style (IEEE recommended), numbered in order of appearance.

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